

NAME: ARCHANA J

REG NUMBER:192324294

Course Code: CSA0920RC

Course Name: Programming in Java

Slot: C

CO3_Assignment - 2

Q2. Try-Catch-Finally Execution Order Write a Java program that demonstrates the execution order of try, catch, and finally blocks. Deliberately trigger an exception (such as an `ArrayIndexOutOfBoundsException` by accessing an invalid array index) inside the try block, handle it in the catch block, and use the finally block to display a message that always executes regardless of whether an exception occurred, such as "Resource closed" or "Cleanup completed".

Concepts: try-catch-finally, execution order.

CODE:

```
public class TryCatchFinallyDemo {  
    public static void main(String[] args) {  
        int[] arr = {10, 20, 30};  
        try {  
            System.out.println("Inside try block");  
            System.out.println(arr[5]);  
            System.out.println("This line will not execute");  
        }  
        catch (ArrayIndexOutOfBoundsException e) {  
            System.out.println("Inside catch block");  
            System.out.println("Exception: " + e);  
        }  
    }  
}
```

```
finally {  
    System.out.println("Inside finally block");  
    System.out.println("Cleanup completed");  
}  
System.out.println("Program continues after try-catch-finally");  
}  
}
```

OUTPUT:



```
Run TryCatchFinallyDemo x  
"C:\Program Files\Java\jdk-17.0.18\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA  
Inside try block  
Inside catch block  
Exception: java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 3  
Inside finally block  
Cleanup completed  
Program continues after try-catch-finally  
  
Process finished with exit code 0
```